

Customer Data Audit: Week 2

Diagnostic findings, key issues, and normalization requirements for the "customers_test" dataset.

DATASET VOLUME

3,150

Total Rows Analyzed

VALIDATED FIELDS



100% Completeness in
"first_name",
"last_name", and
"phone"
(0 NULLs)

IDENTIFIED ANOMALIES

-  **"380"** Missing Emails
-  **"128"** Duplicate Emails
-  **"54"** City Naming Inconsistencies

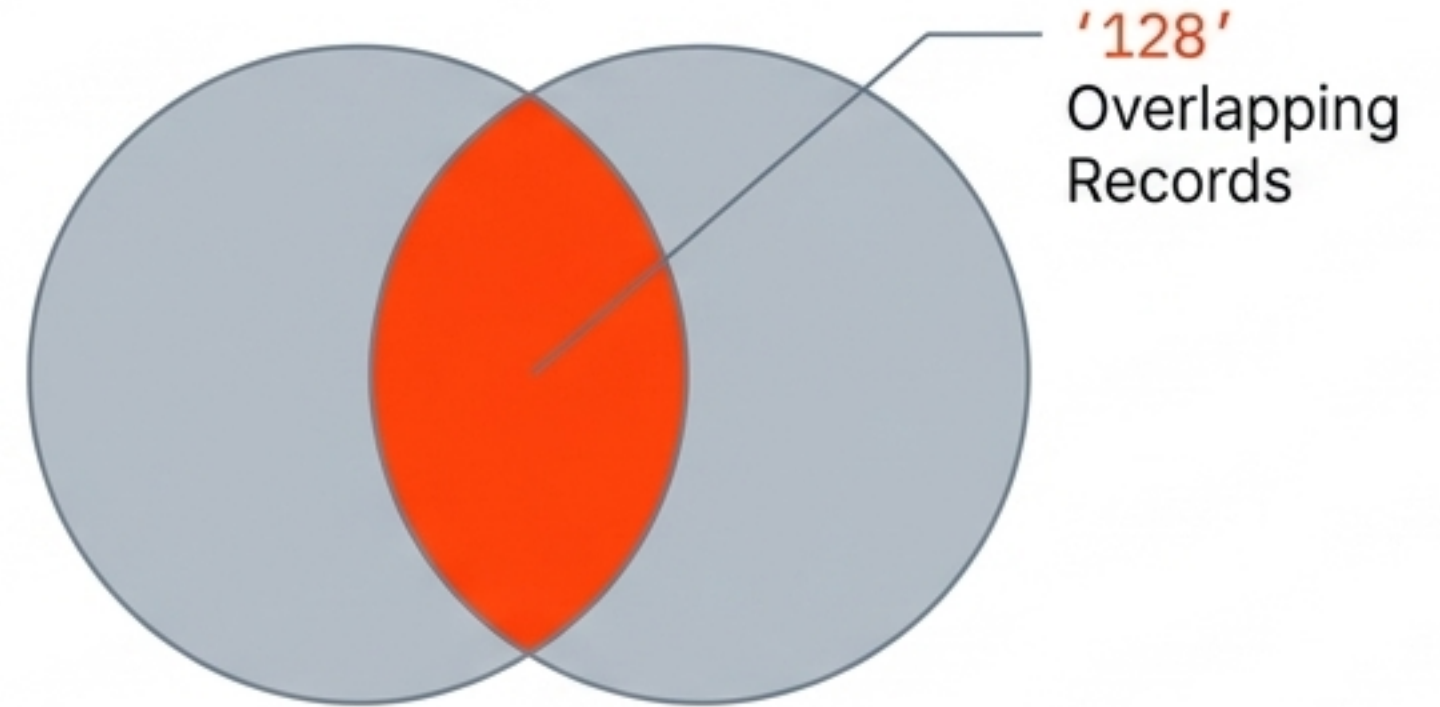
Primary identifiers (Name/Phone) are perfectly populated. Contact routing (Email) and location grouping (City) require immediate programmatic triage.

Completeness Issue



'380 rows' contain NULL 'email' values.

Uniqueness Issue



'128 email addresses' are duplicated across multiple rows.

Strategic action required: Determine if duplicate emails represent repeat customers or system errors, and evaluate if the 380 NULL emails can be backfilled via targeted phone outreach.

The Diagnostic Dashboard

12

True Unique Cities



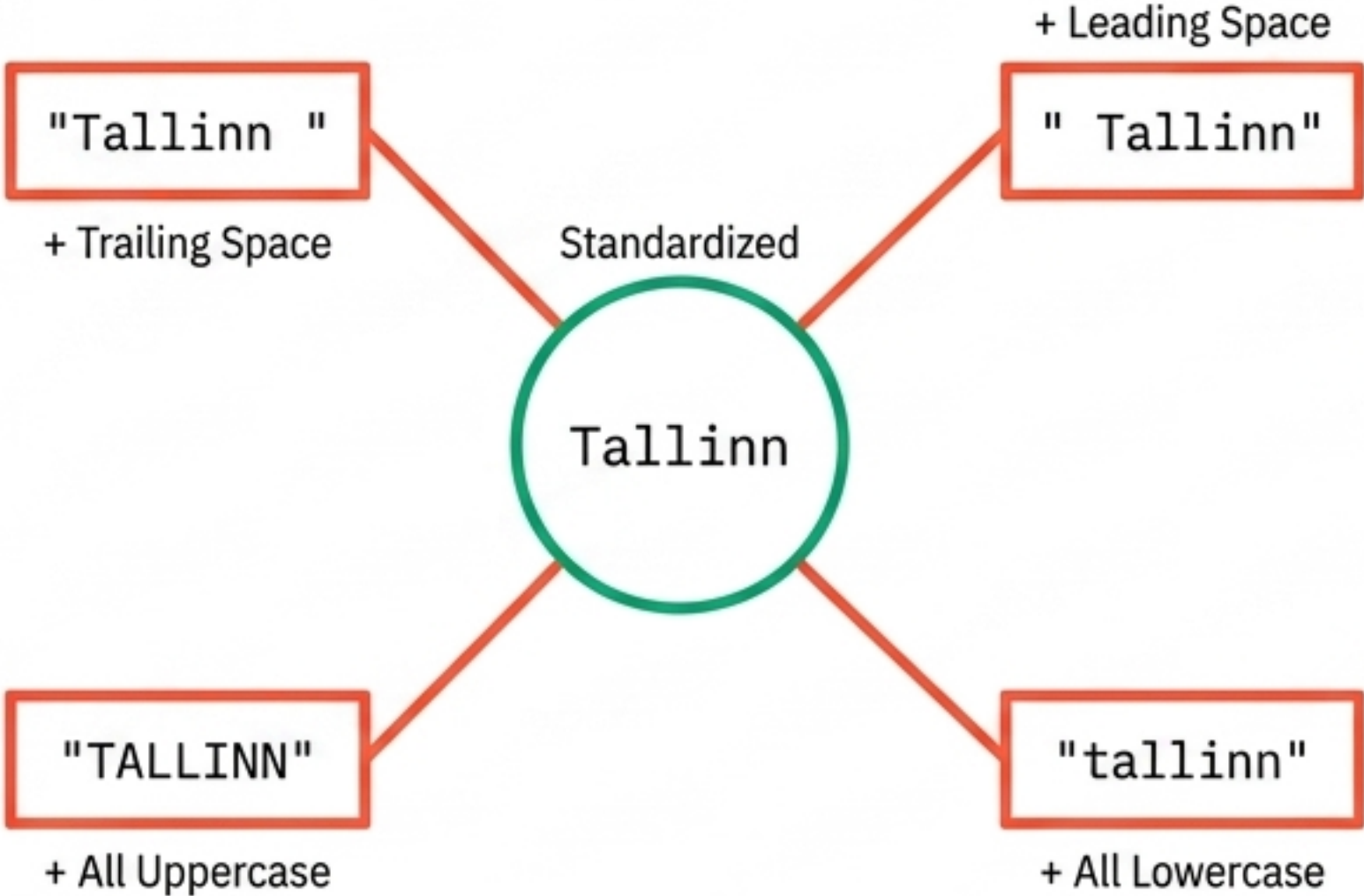
54

Database Entries
("city" column)

Geographical filtering and analytics are effectively impossible in the current state. Without normalization, every true city is fragmenting into multiple ghost entities.

The Diagnostic Dashboard

The Anatomy of the Error



The Impact Table

Remaining Impacted Cities
5 Variations: Võru (Ex: [" Võru", "võru", "Võru", "VÕRU", "Võru"])
5 Variations: Rakvere, Tartu, Viljandi, Kuressaare, Paide, Pärnu
4 Variations: Jõhvi, Narva, Haapsalu (Ex: [" Jõhvi", "jõhvi", "Jõhvi", "JÕHVI"])
2 Variations: Valga (Ex: ["valga", "Valga"])

The formatting variations are highly predictable. They are exclusively combinations of capitalization mismatches and accidental white space.

The Diagnostic Dashboard

Uniqueness

Symptom:

128 Duplicate Emails

Treatment:

Apply `"DISTINCT"` or `group-by` logic to merge/purge duplicate rows based on the most recent system timestamp.

Completeness

Symptom:

380 NULL Emails

Treatment:

Flag these specific rows as `"Phone-Only Contact"` and investigate upstream data capture loss.

Consistency

Symptom:

54 City Spellings for 12 Cities

Treatment:

Apply `"TRIM()"` to remove spaces, followed by `PROPER()` or `LOWER()` to standardize capitalization globally across the column.

Applying these three programmatic transformation rules will yield a 100% clean, 3,150-row dataset ready for production use.